

Harik Sodhi

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EDUCATION

University of Oxford | MEng Engineering Science **2025-2029**

- *Lab project:* Implemented RK4 integrator for rocket landing dynamics (thrust, drag, gravity); designed proportional controller and optimised 1000-dimensional velocity-height profile for fuel efficiency using constrained optimisation via penalty function (MATLAB Optimization Toolbox), with robustness testing across perturbed initial conditions

EXPERIENCE

Oxford Alpha Fund **Oxford**
Director *January 2026 - Present*

- Selected as a Quant Analyst (top ~10% of applicants) via competitive research project and presentation process
- Investigated volatility risk premium in short-dated BTC options; identified statistically significant day-of-week effects (Friday $p = 0.008$, Sunday $p < 0.001$) and constructed systematic straddle strategies achieving Sharpe 1.18 and 70% win rate [github repo](#)
- Developed automated tooling to generate sponsor-ready CV books from member CVs, enabling rapid updates during recruitment cycles
- Evaluated applications and prepared candidates for internship interviews

Optiver Trading Academy **October 2026 - November 2026**
Optiver *Oxford*

- Selected for Optiver's competitive Trading Academy program as a first-year
- Traded ETFs and futures on Optiver's proprietary simulation platform, practising market making, delta hedging, and risk management under realistic market conditions, finishing with a positive P&L

AWARDS & HONOURS

Silver Medal, International Physics Olympiad (49th/420 competitors from 89 countries; highest rank in Western Europe)	2025
UK International Physics Olympiad Team Member	2025
Purple Comet Mathematics Competition — UK National Champion	2024
Winner, Berkeley Math Tournament Online Power Round	2024
BMO1 Distinction	2024

PROJECTS

Market Making Simulator

- Modelled market making against optimal, suboptimal, and insider trader agents; analysed how player-type composition affects spreads and profitability [github repo](#)

DesmosArt

- Built a Python tool converting arbitrary drawings into parametric equations renderable on Desmos, using image processing and curve fitting, animating the drawing process by plotting curves sequentially [github repo](#)